

Unit 4, Lesson 8: Applying Exponent Laws to Algebraic Expressions – Part 2

Benchmark

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Learning Target

- ☐ I can write an equivalent expression by applying quotient of powers or power of a quotient laws of exponents.

4.8.1 Warm-Up: Notice and Wonder

1. The world's largest burger belongs to Mallie's Sports Grill and Bar in Detroit and it costs \$8,000.



- a. What do you wonder about the world's largest burger?
- b. Write a math problem that could be solved using the world's largest hamburger.

4.8.2 Guided Instruction: Equivalent Algebraic Expressions Involving Division

1. For each expression, rewrite the expression in expanded form. Then, write an equivalent algebraic expression using the fewest possible variable bases.

Given Expression	Expanded Form	Equivalent Expression
$\frac{z^8}{z^6}$		
$\frac{x^4y^7}{xy^5}$		

2. Show how each of the equivalent expressions could have been generated by applying the quotient of powers law of exponents.

Given Expression	Apply Quotient of Powers Law of Exponents	Equivalent Expression
$\frac{z^8}{z^6}$		
$\frac{x^4y^7}{xy^5}$		

The quotient of powers law of exponents can be applied to generate equivalent expressions involving division with algebraic expressions.

When dividing algebraic expressions, the coefficients are divided as well. In both expressions above, the coefficients of the divisor and the dividend are both 1, so the coefficient of the resulting expression is also 1 because $1 \div 1 = 1$.

3. Consider the expression $\frac{16x^3y^6}{8xy^3}$. Generate an equivalent expression using division.

$$\frac{16x^3y^6}{8xy^3}$$
$$= \text{ } x \text{ } y \text{ }$$

4. The coefficients will not always be integers. Consider the value of $3 \div 12$.

a. Express the value of $3 \div 12$ as both a fraction and a decimal.

Fraction	
Decimal	

b. Generate an equivalent expression to $-3m^6p^5 \div 12m^4p^2$.

5. Consider the expressions in the table.
- a. For each expression, rewrite the expression in expanded form. Then, write an equivalent algebraic expression using the fewest possible variable bases.

Given Expression	Expanded Form	Equivalent Expression
$\left(\frac{p}{n}\right)^2$		
$\left(\frac{q}{-r}\right)^5$		
$\left(\frac{-2p}{x}\right)^3$		

- b. Complete the statements.

When a quotient is raised to a power, the

- power of a quotient
 - power of a product
 - quotient of a power

law of exponents can be applied by raising both the

dividend and divisor to the same power. If either the dividend or divisor includes a

product, the

- power of a quotient
 - power of a product
 - quotient of a power

law of exponents can also be applied. For

example, $\left(\frac{-2p}{x}\right)^3$ can be expressed $\frac{(-2)^3p^3}{x^3}$, which is equivalent to $\frac{\boxed{}p^3}{x^3}$.

4.8.3 Try It: Equivalent Algebraic Expressions Involving Division

1. Match the expressions in the column on the left to an equivalent expression in the column on the right by drawing arrows.

A. $\frac{-3a^4b^6}{-27a^3b}$

I. $-9a^5b^2$

B. $\frac{27a^4b^6}{3ab^2}$

II. $-\frac{1}{9}a^4b^2$

C. $\frac{27a^4b^4}{-3ab^4}$

III. $9a^3b^4$

D. $\frac{-27a^6b^4}{3a^6b^4}$

IV. $\frac{1}{9}a^3$

E. $\frac{27a^3b^4}{-3a^3b}$

V. -9

F. $\frac{3a^6b^4}{-27a^3b}$

VI. $\frac{1}{9}a^2b^5$

G. $\frac{-3a^7b^9}{-27a^4b^3}$

VII. $-9b^2$

4.8.4 Your Turn!

1. For each expression, apply exponent laws to generate an equivalent expression with the fewest factors possible.

a. $\frac{\pi^9}{\pi^6} = \underline{\hspace{2cm}}$

b. $\frac{5x^4y^7}{15xy^3} = \underline{\hspace{2cm}}$

c. $\frac{-21c^3d^6}{7c^4d^3} = \underline{\hspace{2cm}}$

d. $\left(\frac{x}{2}\right)^2 = \underline{\hspace{2cm}}$

e. $\left(\frac{f}{g}\right)^3 = \underline{\hspace{2cm}}$

f. $\frac{a^7b^9}{a^4b} = \underline{\hspace{2cm}}$

4.8.5 Wrap-Up: Error Analysis

1. Luis rewrote the expression $\frac{-8q^4r^6}{2q^3r^2}$, but he made an error and ended up with an expression that is not equivalent. His work is shown.

$$\frac{-8q^4r^6}{2q^3r^2} = -4q^2r^4$$

- a. What error did Luis make?
- b. Write a brief note to Luis explaining how to avoid making the same error in the future. Be sure to include what he did correctly in your note.